

In 2014 a study was conducted by a PhD candidate at Georgia Southern University examining the correlation between swimming ability, gender, and race. Data from that study included the following information in a two-way frequency table. Use this information to answer questions 1-6.

Ability	Male	Female	Total
Cannot Swim	16.07%	18.50%	34.57%
Can Swim a Little	19.14%	17.11%	36.25%
Can Swim Efficiently	15.37%	13.81%	29.18%
Total	50.58%	49.42%	100.00%

1. Interpret the joint relative frequency females who can swim a little.
17.11% of people surveyed were females who can swim a little.
2. Interpret the value of 34.57% in context of the data in the table.
34.57% of those surveyed cannot swim.
3. Interpret the value of 13.81% in context of the data in the table.
13.81% surveyed were females who can swim efficiently.
4. Interpret the value of $\frac{36.25\%}{100.00\%}$ in context of the data in the table.
Of those surveyed, people that can swim a little.
5. Interpret the value of $\frac{19.14\%}{50.58\%}$ in context of the data in the table.
The percentage of boys that can swim a little
6. What percentage of females can swim efficiently based on the data in the table?
 $\frac{13.81\%}{49.42\%} = 27.94\%$
7. What percentage of those who can swim efficiently are males?
 $\frac{15.37\%}{29.18\%} = 52.67\%$
8. Can you make any associations between gender and swimming ability based on the data? Explain.

No association can be made because the row conditional frequencies are all near 50% and all column conditional frequencies are near 33.3.%.

In the 2014 Georgia Southern University study the correlation between swimming ability and classification in school was also studied. Data from that study included the following information:

- 36.15% of all students were Sophomores.
- 24.34% of all students were Seniors.
- 36.27% of all students can swim a little.
- Of all students, 29.18% swim efficiently.
- 41.02% of seniors can swim a little.
- 7.91% of students that cannot swim are juniors.
- Of the students that cannot swim, 19.54% are sophomores.
- Of the seniors, 23.83% cannot swim.
- 23.75% of students that swim efficiently are juniors.
- Of those students that can swim efficiently, 19.72% are freshmen.
- 1.92% of students were freshmen that can swim a little.

Use this information to answer questions 9-10.

9. Based on the information above create a two-way relative frequency table. Round to the nearest hundredth.

Ability	Freshman	Sophomore	Junior	Senior	Total
Cannot Swim	19.27%	6.75%	2.73%	5.80%	34.55%
Can Swim a Little	1.92%	21.46%	2.91%	9.98%	36.27%
Can Swim Efficiently	5.75%	7.94%	6.93%	8.56%	29.18%
Total	26.94%	36.15%	12.57%	24.34%	100%

10. Does there appear to be an association between the abilities to swim and the class of students in school?

The relative frequencies in the table do appear to show a weak association between the grade a student is in and their ability to swim.